SIMATS ENGINEERING

ASSESSMENT TOOL II — Visualization Analysis Report

Course: Data Handling and Visualization	**Code: DSA06**	**CO: CO5**
Duration: 180 min	**Total Marks: 100**	**Reg No :** 192524225 **Name**: VALLAPU MOHAN KUMAR

COURSE OUTCOME COVERED

CO5: Analyze and evaluate visualization choices to communicate patterns, comparisons, relationships, and potential misleading effects through appropriate visualization methods. (BL4)

Activity Tool : Visualization Analysis Report

Weightage: 20%

Code of Conduct:

I VALLAPU MOHAN KUMAR (192524225) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V. Mohan Kumar

Assessment Criteria

Criteria	Visualization Selection and Understanding 25%	Visualization Analysis and Interpretation 25%	Application of Visualization Principles 20%	Organization and Presentation 20%	Accuracy 10%	Clarity 5%	Total
Q1							
Q2							
Q3							
Q4							
Q5							

Visualization Analysis Report – Assessment Tool

R Programming Practical Questions with Datasets

Instructions: Answer all five questions. Use R programming to construct the required visualizations, analyze the patterns, compare visualization choices where requested, and identify potential visualization pitfalls. Use the given data exactly unless a transformation is explicitly requested.

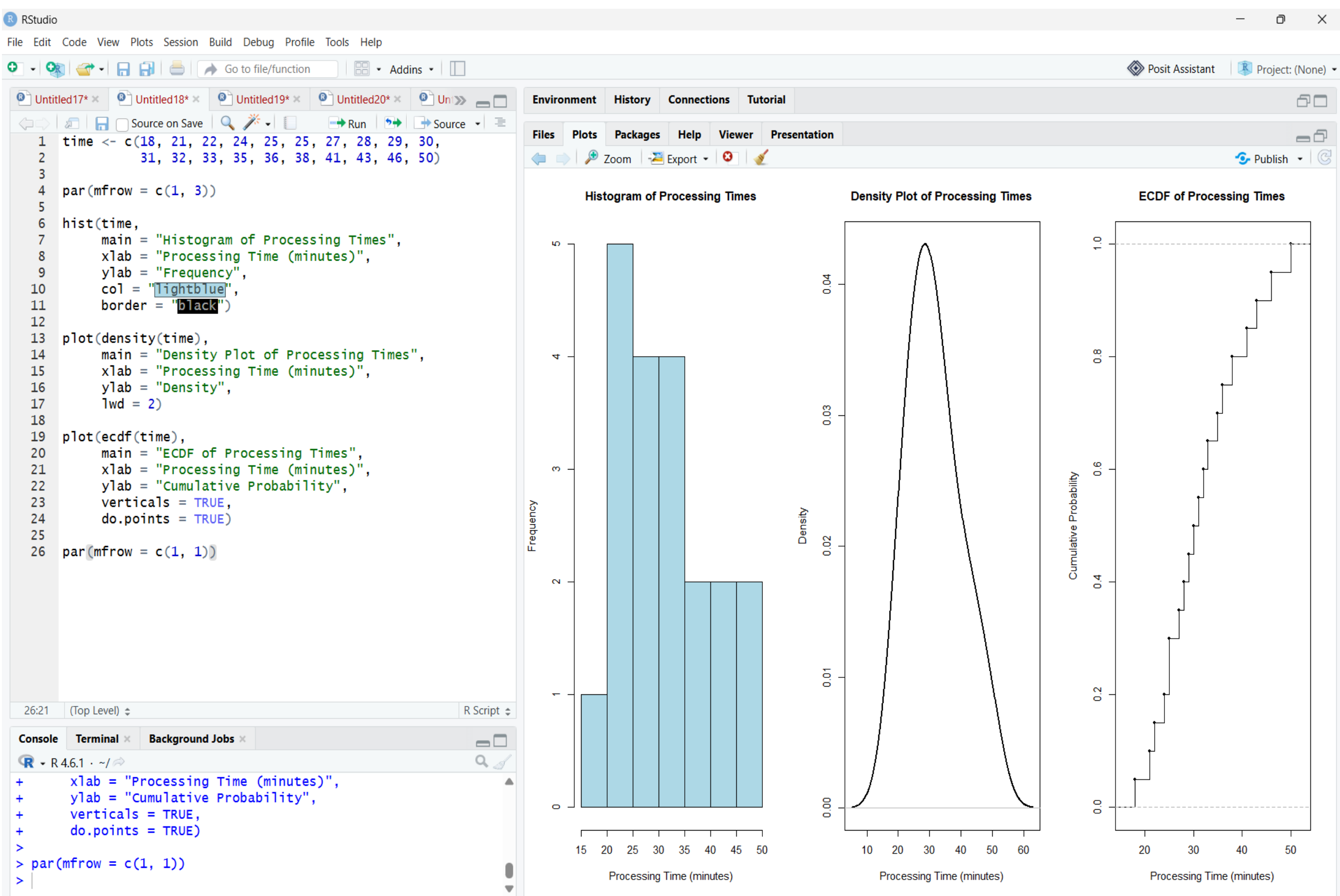
Q1. Comparing Distributions – Histogram, Density and ECDF

A quality-control team records the following processing times (in minutes) for 20 randomly selected tasks. Using R, create a histogram and a density plot for the data. Also construct an empirical cumulative distribution function (ECDF). Compare the three visualizations and explain which aspects of the distribution—center, spread, skewness, and cumulative behavior—are communicated most clearly by each.

Data: Time = 18, 21, 22, 24, 25, 25, 27, 28, 29, 30, 31, 32, 33, 35, 36, 38, 41, 43, 46, 50 *Expected R tasks: histogram,*

density estimation, ECDF, appropriate labels/scales, comparison and interpretation

Answer:



Interpretation

- Histogram: Shows how frequently processing times occur in different intervals.
- Density plot: Provides a smooth representation of the distribution and makes the overall shape easier to observe.
- ECDF: Shows the proportion of tasks completed at or below each processing time.
- The center of the data is around 30–32 minutes.
- The observations extend from 18 to 50 minutes, indicating moderate spread.
- The higher values toward 41–50 minutes indicate a slight right skew.
- The histogram is best for frequency, the density plot is best for distribution shape, and the ECDF is best for cumulative behavior.

. **Q2. Q-Q Plot and Normality Analysis**

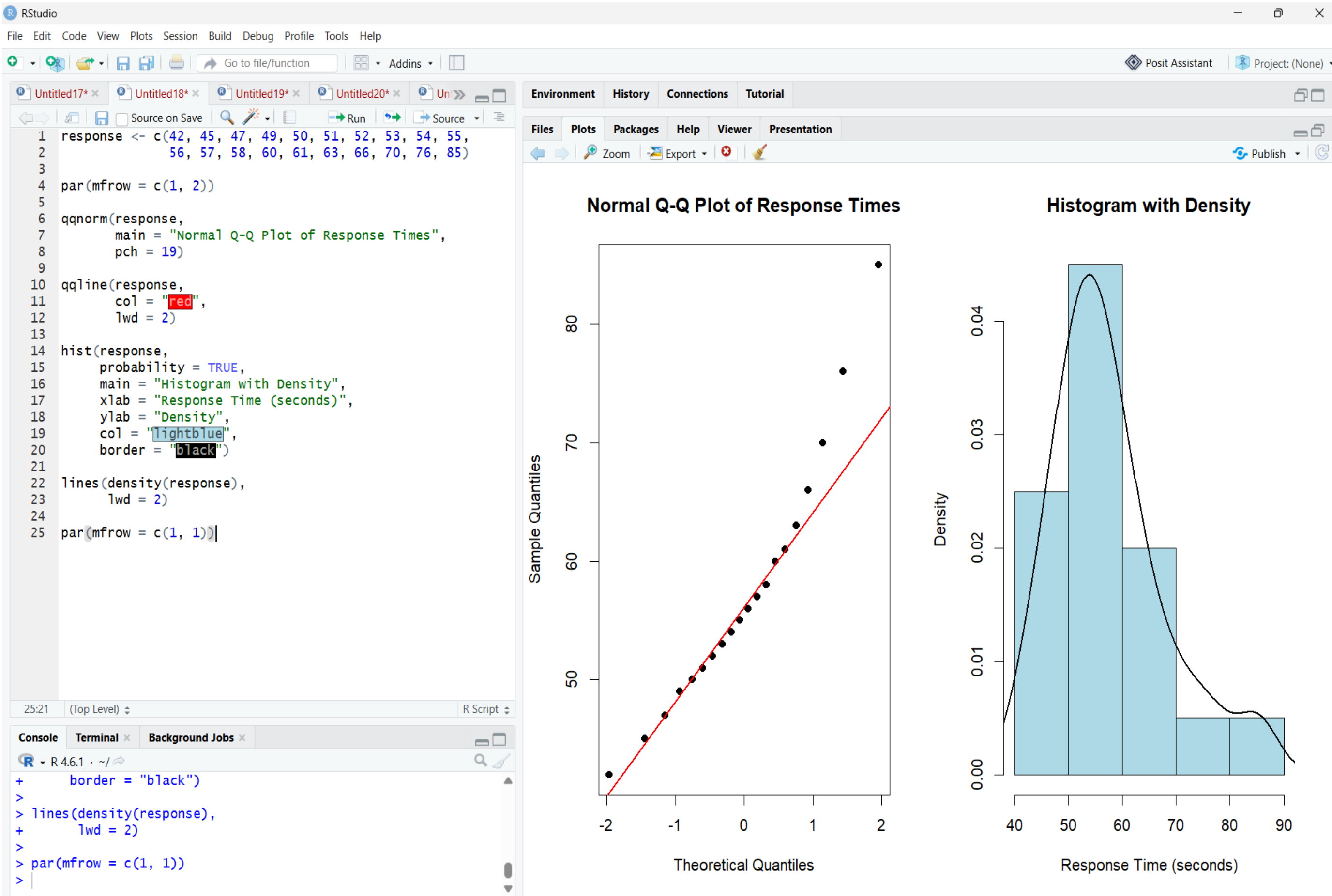
A researcher measures the following response times (in seconds). Use R to create a normal Q-Q plot and a histogram/density visualization. Assess whether the observations are reasonably consistent with a normal distribution. Identify any departures from the reference line and explain what they indicate about the distribution.

Data: Response = 42, 45, 47, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 60, 61, 63, 66, 70, 76, 85

Expected R tasks: qqnorm()/qqline() or ggplot2 Q-Q plot, histogram/density plot, interpretation of deviations and outliers.

Answer:

The response-time data contains 20 observations. The task asks for a normal Q-Q plot and histogram/density visualization and requires assessment of departures from normality.



Interpretation

- Most observations are reasonably close to the Q-Q reference line.
- The extreme upper observation, 85 seconds, deviates noticeably from the reference line.
- The value 76 seconds also contributes to the upper-tail departure.
- The histogram shows a concentration of observations around 50–60 seconds.
- The distribution has a slight right-tail extension.
- Therefore, the data are approximately normal in the central region but show some departure from normality in the upper tail.
- The Q-Q plot is particularly useful for identifying these tail deviations.

Q3. Visualizing Many Distributions at Once

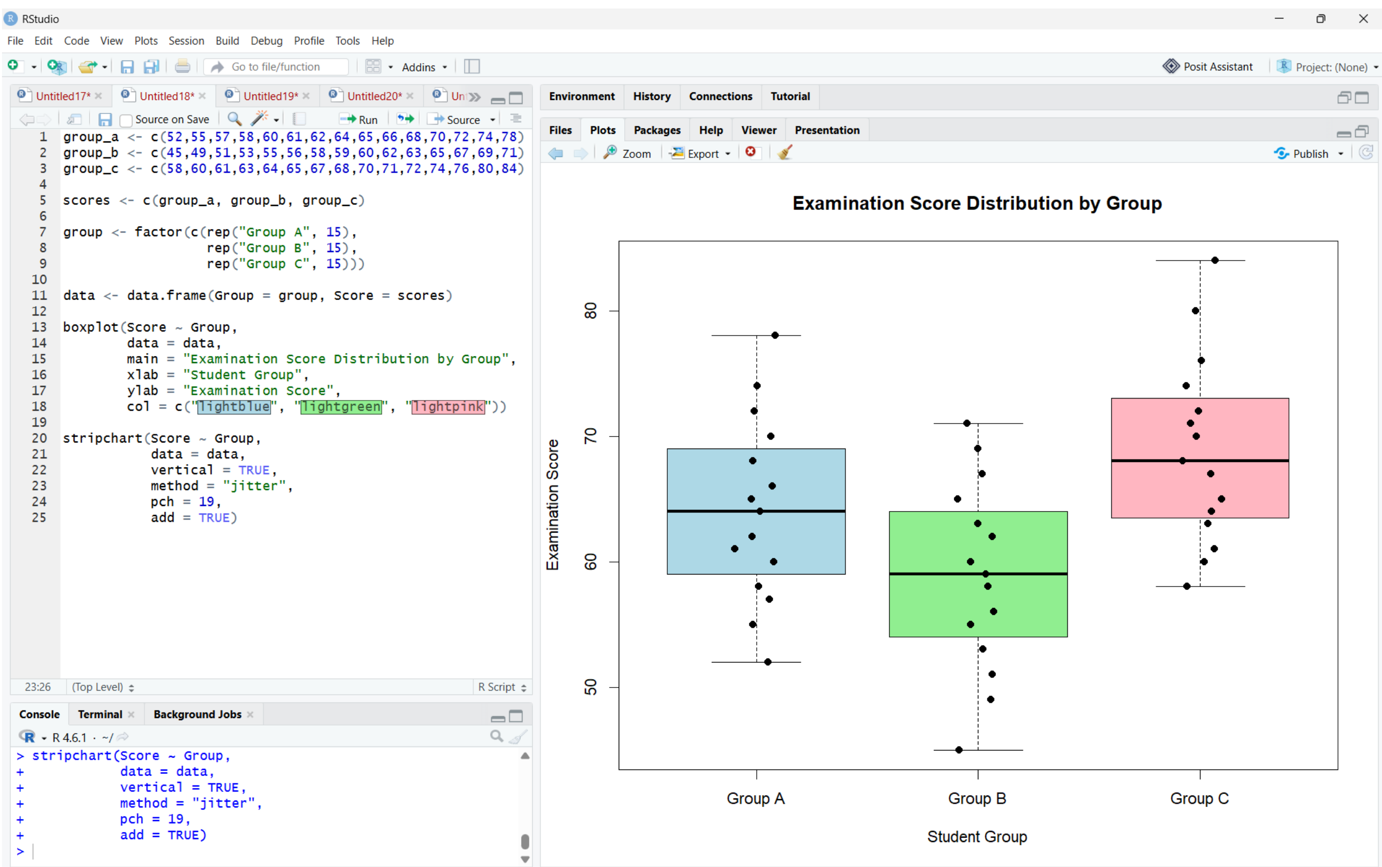
Three groups of students have examination scores recorded below. Create a suitable visualization that compares the distributions of all three groups at once. Use boxplots and/or violin plots, and optionally overlay individual observations. Analyze differences in center, spread, and possible outliers. Explain why the selected visualization is more informative than displaying three separate plots.

Group A = 52,55,57,58,60,61,62,64,65,66,68,70,72,74,78;
Group B = 45,49,51,53,55,56,58,59,60,62,63,65,67,69,71;
Group C = 58,60,61,63,64,65,67,68,70,71,72,74,76,80,84

Expected R tasks: data reshaping, grouped boxplot/violin plot, optional jitter, comparison of distributions, interpretation.

Answer:

The question provides examination scores for Groups A, B, and C and asks for a visualization comparing their distributions, including center, spread, and possible outliers.



Interpretation

- **Group A** has a relatively high center, with scores concentrated around the mid-60s.
- **Group B** has the lowest center and includes scores ranging from 45 to 71.

- **Group C** has the highest center and extends up to 84.
- The boxplot makes it easy to compare the **median, spread, and extreme values** of all three groups.
- The overlaid individual observations show the actual student scores.
- A combined boxplot is more informative than three separate plots because the groups share the **same scale and visual reference**.
- The visualization allows direct comparison of the groups without switching between separate figures.

Q4. Relationships and Multivariate Visualization

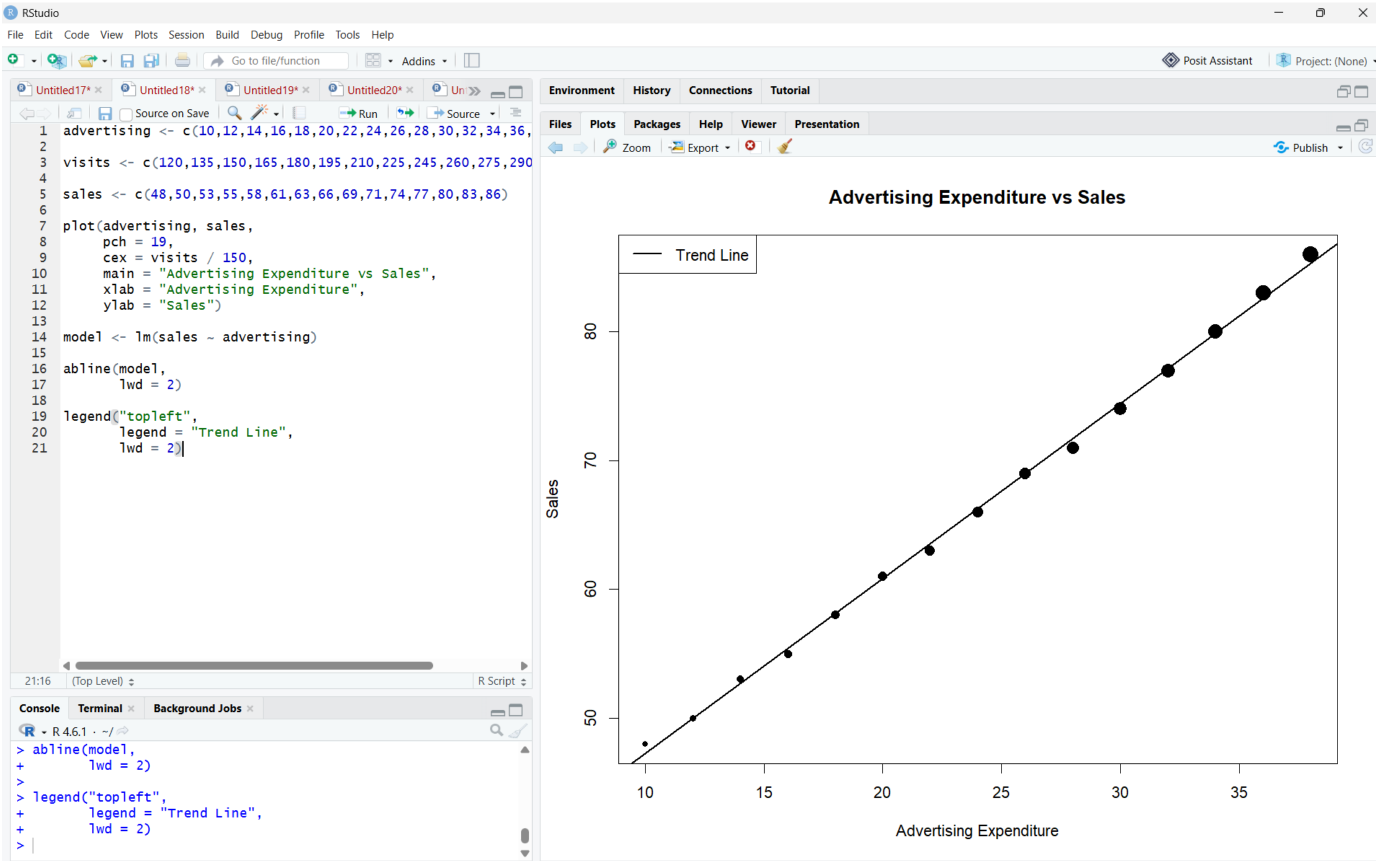
A study records advertising expenditure, website visits, and sales for 15 observations. Create a scatterplot of advertising expenditure versus sales and use an appropriate visual encoding to incorporate website visits. Add a fitted trend line. Analyze the strength and direction of the relationship and explain whether the additional variable improves the interpretation or risks cluttering the visualization.

Advertising = 10,12,14,16,18,20,22,24,26,28,30,32,34,36,38;
Visits = 120,135,150,165,180,195,210,225,245,260,275,290,305,320,340;
Sales = 48,50,53,55,58,61,63,66,69,71,74,77,80,83,86

Expected R tasks: scatterplot, trend line, third-variable encoding, appropriate legend/scale, relationship analysis and critique.

Answer:

The data contains advertising expenditure, website visits, and sales for 15 observations. The task requires a scatterplot, incorporation of website visits, a fitted trend line, and relationship analysis.



Interpretation

- The scatterplot shows a **strong positive relationship** between advertising expenditure and sales.
- As advertising expenditure increases, sales also increase.
- The fitted trend line confirms the positive direction of the relationship.
- Website visits are incorporated through the **size of the points**.
- Larger points represent higher numbers of website visits.
- This third-variable encoding provides additional information without requiring a separate graph.
- However, using too many visual encodings can create clutter, especially when several variables have similar visual importance.

- In this dataset, the additional variable improves interpretation because website visits generally increase along with advertising expenditure.

Q5. Misleading Visualization and Redesign

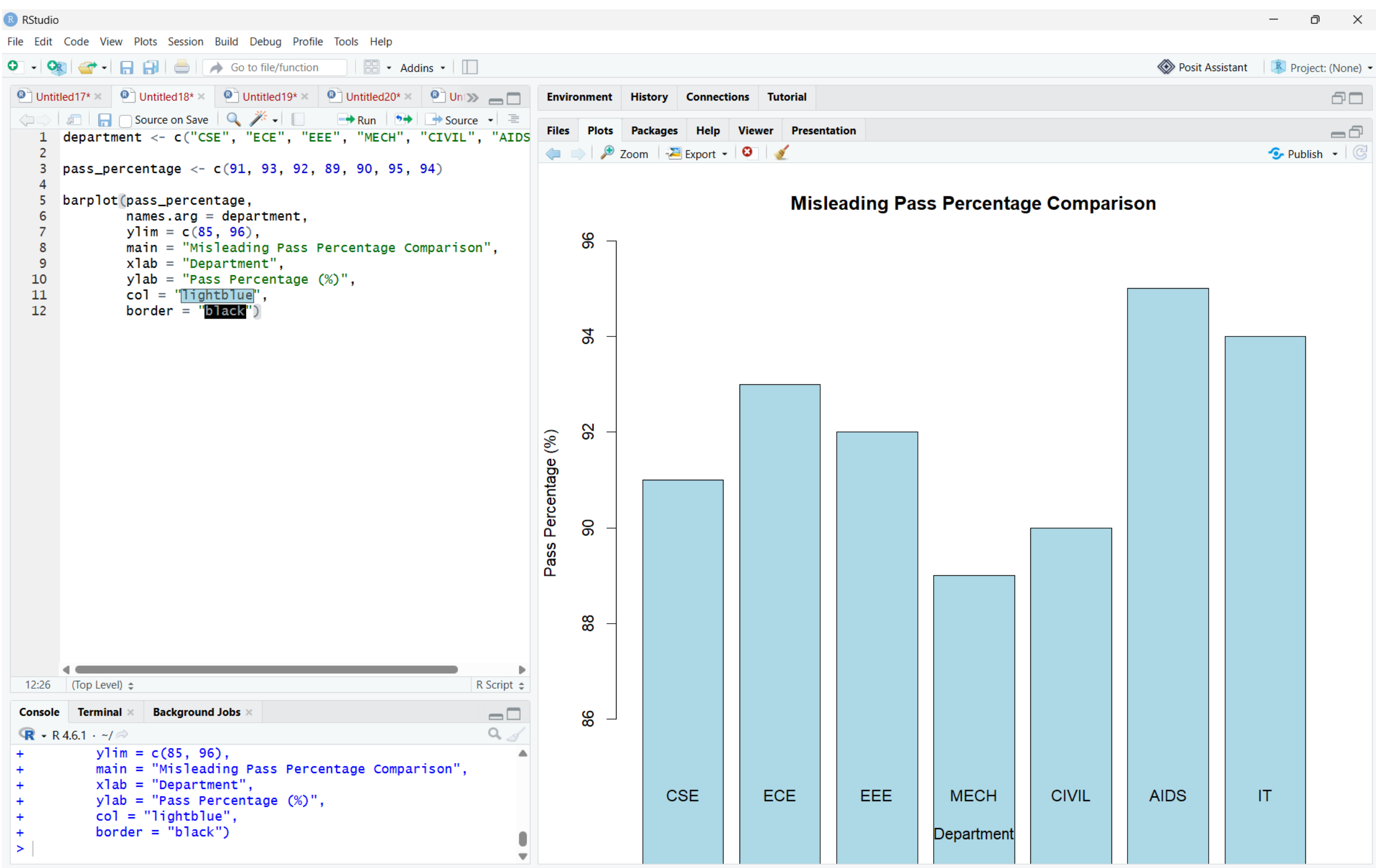
A college reports the following pass percentages for seven departments. Create two visualizations: (1) a misleading version using a truncated y-axis and an inappropriate visual emphasis, and (2) a redesigned version using an appropriate baseline, ordered categories, and clear labels. Compare the two plots and explain how the first visualization can exaggerate differences and why the redesigned visualization communicates the data more honestly.

Department = CSE, ECE, EEE, MECH, CIVIL, AIDS, IT;
Pass_Percentage = 91,93,92,89,90,95,94

Expected R tasks: bar chart with controlled y-axis, redesigned bar chart, ordering, labels, misleading-axis critique, accessibility and communication analysis.

Answer:

The question provides pass percentages for seven departments and asks for a misleading visualization using a truncated y-axis followed by a redesigned visualization with an appropriate baseline, ordered categories, and clear labels.



Why it is misleading

The y-axis starts at **85% instead of 0%**.

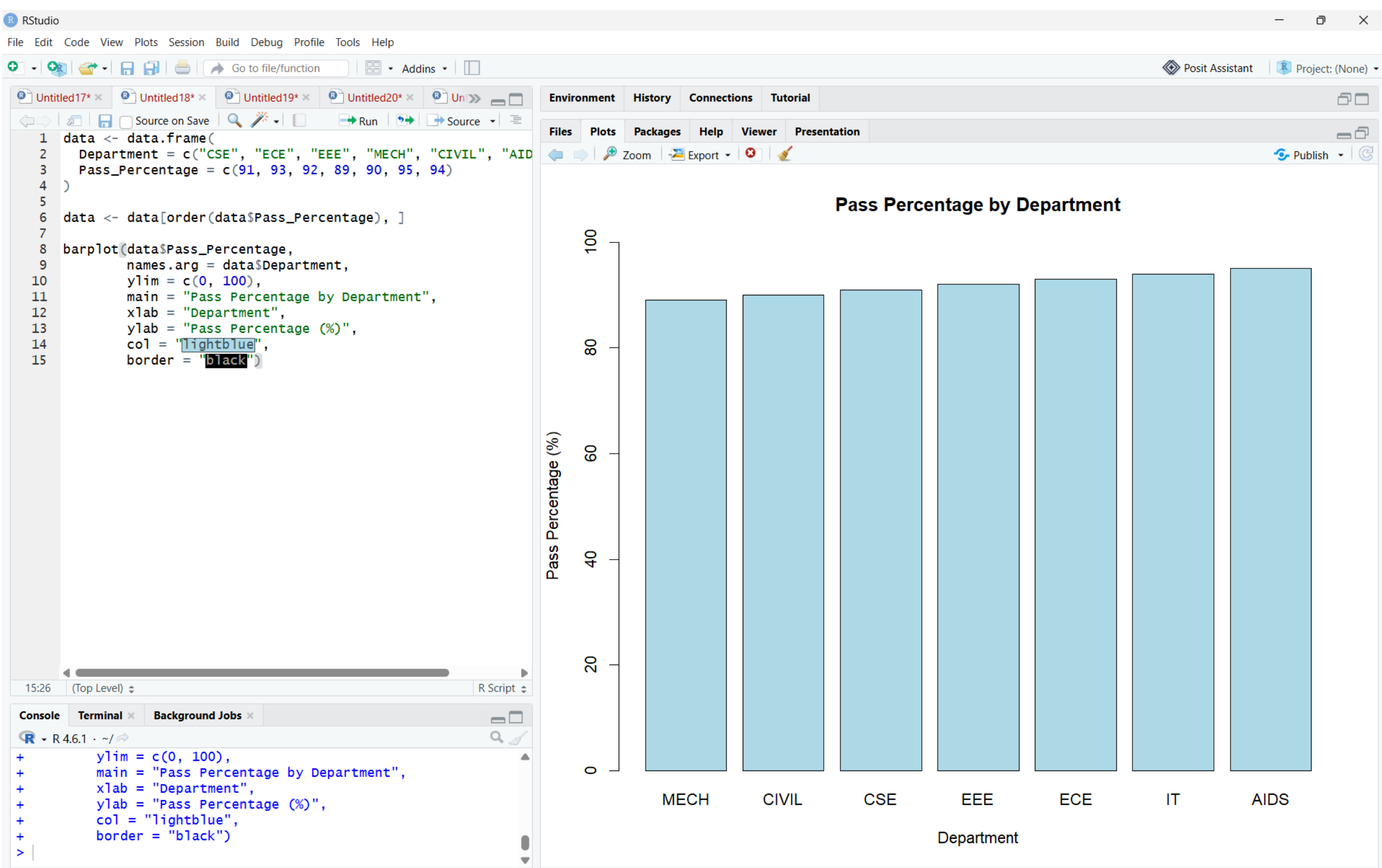
Therefore, small differences between departments can appear much larger than they actually are.

For example:

- CIVIL = 90%
- AIDS = 95%

The actual difference is only **5 percentage points**, but a truncated axis visually exaggerates this difference.

Part B — Redesigned Visualization



Interpretation

- The redesigned graph starts the y-axis at **0%**, providing an honest visual comparison.
- Departments are ordered according to their pass percentages.
- **AIDS** has the highest pass percentage at **95%**.
- **MECH** has the lowest pass percentage at **89%**.
- The difference between the highest and lowest departments is only **6 percentage points**.
- The redesigned graph avoids exaggerating differences.
- Clear labels and an appropriate baseline make the visualization easier to understand and more trustworthy.